

WRRRA M Downstream Integrated Model and Verification

From channel selection to energy pressure and proper time spectra

Wonsik Choi and Jeongin Choi

Correspondence: Wonsik Choi, Independent Researcher, Seoul, Republic of Korea

ORCID 0009-0001-4263-9772 · janefather@gmail.com

WRRRA M 1.0 final consolidated edition of downstream 0.1 through 0.12

Abstract

We integrate the executable downstream development of WRRRA M from 0.1 to 0.12 into one account of channel selection, retained address information, physical energy and pressure, geometric responses, and proper-time spectra. The frozen WRRRA Core 1.0 and Minimal Computation Cosmology 2.3.2 provide the execution architecture. Verified physical values calibrate the selected our-universe construction. A common carrier connects conditional sixteen-channel algebra and calibrated filter selection to positive state loads. A common address measure supplies a 5% / 26.8% / 68.2% partition, while explicit SI response coefficients preserve the separate inherited 4.93% / 26.5% / 68.57% energy calibration. The volume derivative of the same energy gives pressure and the adopted homogeneous response gives deceleration. Electron mode 23 fixes the energy unit used in finite boundary spectra and worldline proper-time updates. Clean replay reproduces 96 component check groups and 37 result and bilingual source files exactly. The old slow nonuniform expansion schedule fails identification with the SI Hamiltonian phase. Physical measurement outcomes, post-observation states and physical records remain unimplemented. The resulting 1.0 is the final downstream consolidated research edition with an explicit executable scope and closure ledger.

Keywords minimal computation common carrier dimension filter phenotype information load proper time WRRRA

1 Introduction and evaluation rule

WRRRA means Wonsik Reality Renderer Architecture. It connects the preparation, transport, update and physical readout of a state under a minimum sufficient execution structure. Minimal computation concerns the information and reusable rules sufficient to sustain the selected execution, including required reserve and robustness. It does not impose a unique numerical exponent or a unique universe. A common carrier supplies shared transport and principal kinematics; internal representations and readout rules can still produce different phenotypes.

The question of this paper is how the same state can be followed from a structural channel model and retained address ledger into physical loads, pressure, geometry and clocks. This connects the matter-fraction manuscript r9 and extension manuscript r8 [1,2] to the reviewed downstream releases [3-5]. The contribution is the implemented transformation and its connected explanation. Calibration to known values and their faithful reproduction are model achievements. Numerical novelty, calibration-free constants and a new independent prediction are not requirements for this contribution. After the model and calibration are fixed, a calculated unmeasured quantity is a conditional WRRRA prediction with its inputs and falsifiers.

Every assessment follows verification inputs, WRRRA-specific transformation, outputs and falsification conditions. The inputs include established measurements, exact unit definitions and disclosed

construction choices. The transformation must actually execute the claimed connection. An external theory becomes an integrated calculation only where its equations are run on the same declared state and load. Representation algebra, homogeneous expansion and finite local response are therefore stated at their executed scope; reference to a theory alone does not establish its integration.

Downstream development ends at 0.12. Version 1.0 consolidates 0.1 through 0.12; upstream studies retain their separate scope. The endpoint correction withdraws the automatic prospective downstream stages 0.13 through 0.17. It preserves historical calculations and does not convert unresolved connections into completed results [6]. WRRRA M remains a foundational toy model of one nonprivileged universe, calibrated to the adopted our-universe inputs.

2 Inputs terminology and development scope

Core 1.0 and MCC 2.3.2 are frozen [7,8]. The five physical calibration anchors are supplied sequentially as G , H_0 , phenotype energy fraction, clustering energy fraction and electron rest energy. Exact SI references c , h and the electronvolt conversion set units. The inherited H_0 benchmark is a Planck base Λ CDM calibration choice; it is not a model-independent determination from the downstream code. The two address-fraction targets remain additional arithmetic inputs.

Table 1 Calibration ledger

Quantity	Adopted value	Role
G	$6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	Gravitational normalization
H_0	$67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Homogeneous density benchmark
Physical fractions ϕ D R	0.0493 / 0.265 / 0.6857	Inherited SI energy targets
Electron rest energy E_e	510998.95069 eV	Electron-mode energy anchor
c	$299792458 \text{ m s}^{-1}$	Exact SI reference
h	$6.62607015 \times 10^{-34} \text{ J s}$	Exact SI reference
1 eV	$1.602176634 \times 10^{-19} \text{ J}$	Exact conversion
Arithmetic fractions ϕ D R	0.05 / 0.268 / 0.682	Joint address partition

Addresses, channel labels and physical coordinates are different objects. An address n is not automatically a particle label, spatial length or bit count. The 128-site carrier grid is not the sixteen-channel inventory. Information load is a positive state-dependent expectation entering the energy response; it is not Shannon information, entropy or the number of channels. A physical record retains a generated observation outcome. A calculation log, input provenance and a unitary overlap are separate objects.

In the matter-fraction paper, retained Actual means the admitted residue $\phi + D$. In the complete downstream ledger, the accounted state also includes the returned provenance and its background response R . We use resident Actual for the former and complete accounted ledger for the latter. R denotes rejection and return in the address partition and the associated background response in the physical map. Physical records are written as R to avoid conflating these meanings. The earlier f_c and f_b correspond to D and R in the physical ledger.

Table 2 Downstream sequence and its final scope

Stage	Completed result	Condition retained in 1.0
0.1	Fifteen-channel color multiplicities	Predimensional interpretation and Weyl realization are assumptions
0.2	Sixteen-channel placement hypercharge and anomalies	Conditional pairing and normalization
0.3	Four-filter selection and stability theorem	Positive gaps and nonzero winner support
0.4	Common-carrier metric and neutral-channel lift	Positive transport norm and shared principal geometry
0.5	Finite local twist response and mass-mode example	Symmetric carrier gives zero selection gaps
0.6	Homogeneous energy pressure twist and expansion	Chosen finite geometry and constitutive clock
0.7	Energy-weighted Actual and phenotype measure	Fractions are load shares rather than measurement probabilities
0.8	Calibrated carrier-response filter selection	Known origin assignment calibrates readout orientation
0.9	Common address partition and upstream interface	Dimensionless admission sequence and declared retention
0.10	SI address energy pressure and conversion work	Product address-carrier state and fixed response shapes
0.11	Sequential five-anchor calibration	Residual response and geometric freedoms
0.12	Proper-time updates finite spectra and SI conversion	Boundaries adopted and slow-clock identification fails

3 Common carrier and conditional particle selection

The starting structural calculation uses two seven-dimensional sectors and one invariant channel. Under the $SU(3)$ color subgroup, each seven branches into a triplet, antitriplet and singlet. Thus the fifteen-channel space contains two triplets, two antitriplets and three singlets, matching the color multiplicities of one minimal Standard Model generation [3]. Seven predimensional degrees are not seven measured spatial directions. The fifteen-channel count alone does not derive weak charges, chirality, masses or generations.

$$C_{15} = 1_0 \oplus 7_A \oplus 7_B, 7 \downarrow SU(3) = 3 \oplus \bar{3} \oplus 1. \quad (1)$$

The neutral extension adds a transported channel. Gauge neutrality means no Standard Model gauge-current readout; it does not mean a zero transport vector. Positive norm, full channel transport and a common principal causal geometry are required. The adopted pairing reorganizes the sixteen channels into (4,2,1) and (anti-4,1,2). Conditional on this pairing and the neutral and invariant channel normalizations, one hypercharge rule produces the known one-generation assignments and the exact anomaly cancellations.

$$C_{16} = C_{15} \oplus 1_N, Y = T_{3R} + \frac{B-L}{2}, Q = T_{3L} + Y. \quad (2)$$

The admissible selection class contains four color-preserving pairings. A positive common-carrier metric defines every compatibility by negative squared response mismatch. The left and right gap

identities then follow by expanding the same squared norm. The target F_DX is the unique maximum precisely when both gaps are positive.

$$\begin{aligned} c(i, j) &= -\|u_i - u_j\|_C^2, \\ \Delta_L &= 2 \operatorname{Re} \langle u_{3_A} - u_{3_B}, u_{1_A} - u_{1_B} \rangle_C, \\ \Delta_R &= 2 \operatorname{Re} \langle u_{3_A} - u_{3_B}, u_{1_N} - u_{1_0} \rangle_C. \end{aligned} \quad (3)$$

Positive-rate replicator flow converges from nonzero winner support. Entrywise score perturbations bounded by ρ preserve the selection when the smaller gap exceeds 4ρ . These are exact conditional statements within the four-filter class, not a proof that all microscopic filters belong to that class.

$$p_f(s) = \frac{p_f(0) e^{\eta C_f s}}{\sum_g p_g(0) e^{\eta C_g s}}, \eta > 0. \quad (4)$$

The symmetric common carrier in 0.5 has full transport rank and zero gaps. This negative result is preserved. In 0.8, origin-specific offsets 0.75 and 1.25 in the auxiliary readout operator give different positive resolvent responses while retaining common principal transport. The fixed known origin assignment calibrates orientation. At the uniform reference the responses are 0.672634137 and 0.825428002, producing $\Delta_L = \Delta_R = 0.04669193039$. F_DX wins all nine disclosed state-volume cases. Zero contrast restores the tie, and reversing relative orientations changes the winning pairing [4].

The selection parameter s and candidate weights are optimizer coordinates and weights. They are not physical time or Born probabilities. One transported neutral channel does not establish a measured sterile neutrino, its occupation or its mass. Three calibrated generations allocate 48 channel slots with normalized generation weights; the allocation preserves one total energy budget. This is channel and charge accounting, with Yukawa interactions, mixing and the full interacting Standard Model outside the calculated scope.

4 Retained addresses and physical readout of the same state

The Physics matter manuscript uses prime factorization to distinguish independent SOURCE addresses from composite relations [1]. The fold index $\Omega(n) - 1$ counts retained factor multiplicity; relation dimension is zero for independent prime addresses and one for composite relations in this model. These are address-structure definitions. A composite can enter through an initially blurred filter while a normalized return filter preserves factors and blocks return to a zero-relation SOURCE. A retained state remains in the return kernel only if later updates preserve that subspace.

$$\begin{aligned} \rho \geq 0, \operatorname{Tr} \rho = 1, A^\dagger A \leq I, B^\dagger B \leq I, \\ q_{res} = \operatorname{Tr} \left[\rho A^\dagger (I - B^\dagger B) A \right], q_R = 1 - q_{res}. \end{aligned} \quad (5)$$

The final renderer pairs address components in binary form and uses the unpaired remainder as the phenotype coupling port. Odd composite residue can activate that readout; even composite residue remains nonphenotypic in the adopted map. Lack of phenotype does not remove positive load. The general two-filter operator formulation and its finite diagonal implementation remain distinct.

The active address range is 2 through 1000000. Address 1 is the multiplicative identity and adopted vacuum reference, outside this active normalization. Excluding it does not by itself set physical vacuum energy to zero. All address sectors use the same normalized measure.

$$Z_N(\alpha) = \sum_{n=2}^N n^{-\alpha}, w_n = \frac{n^{-\alpha}}{Z_N(\alpha)}, \sum_n w_n = 1. \quad (6)$$

With $\alpha = 2$ and complete odd-composite admission, the finite residue is 4.876893353%; its arithmetic limit is 4.876952810%. The joint constant-admission model calibrates $\alpha = 1.8996876950554356$ and $\beta = 0.865457012413696$ to 5% phenotype and 26.8% even-composite load. Return 68.2% follows by closure. The 4.8769% result at $\alpha = 2$ and the separate 26.7942% comparison at $\alpha = 1.9$ cannot be summed across different normalizations.

The phase-driven 0.9 implementation retains the same address target ledger. Four fixed zeta-zero heights, equal coefficients, phase increment 0.1 and $K = 8$ admission steps produce address-dependent admission. Threshold $h = 1.447673170352440$ calibrates the phenotype fraction. Each new admission depletes the pending reservoir, preventing repeated counting.

$$\begin{aligned}\delta_n(k) &= \sum_j c_j \cos[\gamma_j(\log n + \xi k) + \theta_j], \\ a_n(k) &= (1 + e^{h - \delta_n(k)})^{-1}, T_n(K) = 1 - \prod_{k=0}^{K-1} (1 - a_n(k)).\end{aligned}\quad (7)$$

$$e_\varphi = 1_{C_o} T_n, e_D = 1_{C_e}, e_R = 1 - e_\varphi - e_D, f_s = \sum_n w_n e_s(n). \quad (8)$$

The depletion ledger maintains $\varphi + D + S + R = 1$. At the declared recovery boundary, pending S moves into R without being counted twice. Resident Actual is $\varphi + D = 31.8\%$; the complete accounted ledger is 100%. Constant-admission and phase-driven models can fit the same totals while giving different address distributions. Their aggregate agreement does not establish identical microscopic histories or physical seconds.

The SI connection in 0.10 joins the diagonal address state with the carrier state by an explicit product construction. Address responses are positive and constitutive. The reference uses $\lambda_\varphi = 0$, $\lambda_D = 0.25$ and $\lambda_R = 0.1$. Address state and carrier load are then varied without refitting their reference SI coefficients.

$$\begin{aligned}\sigma &= \sum_n w_n |n\rangle \langle n| \otimes \rho, \\ g_s(n) &= 1 + \lambda_s \frac{\log n}{\log N}, \mu_s = \sum_n w_n e_s(n) g_s(n).\end{aligned}\quad (9)$$

This executes an address-to-load map, not a complete microscopic identification of each address with an observed particle. The shared upstream interface supplies state, measure, phase spectrum, admission and retention rules, channel routing and source hashes. General address-carrier entanglement dynamics remain open.

5 One energy function for pressure gravity and expansion

For the carrier, K_c is the positive periodic Laplacian and K_b its disclosed positive noncommuting control. Let $A_\varphi = I$, $A_D = K_c / 2$ and $A_R = K_b / 2$, so each uniform reference expectation is one. The reference volume $V_0 = 1 \text{ m}^3$ is a bookkeeping volume, not the cosmic size.

$$u_{crit} = \frac{3 H_0^2 c^2}{8 \pi G}, \eta_s = \frac{u_{crit} t_s}{\mu_s}, [\eta_s] = J m^{-3}. \quad (10)$$

$$\begin{aligned}\hat{E}(a) &= V_0 \sum_s \eta_s \mu_s a^{v_s} A_s, \\ E_s &= V_0 \eta_s \mu_s a^{v_s} \text{Tr}(\rho A_s), E = \sum_s E_s.\end{aligned}\quad (11)$$

The reference exponents $v_\phi = v_D = 0$ and $v_R = 3$ give pressureless comoving phenotype and clustering energies and a constant background energy density. Pressure is derived from this energy, rather than assigned independently. A five-point volume derivative checks the same expression.

$$V = V_0 a^3, u_s = E_s / V, P_s = - \left(\frac{\partial E_s}{\partial V} \right)_{\rho, w, e, g} = - \frac{v_s}{3} u_s. \quad (12)$$

The inherited homogeneous action executes the Friedmann and acceleration equations on these loads. The global weighted twist reads $D + R$, whereas the inherited local extra-attraction scale reads D . The factor $1/8$ and local response function are disclosed calibrated connections. They are not derived by the arithmetic address partition alone.

$$H^2 = \frac{8\pi G}{3c^2} u, q = \frac{u + 3P}{2u}, \quad (13)$$

$$a_T = c H_0 \sqrt{\frac{u_D}{8u_{crit}}}, \zeta_T \kappa_h^2 = \frac{16\pi G}{c^4} (u_D + u_R).$$

The local response is $g = \frac{g_M}{1 - \exp[-\sqrt{g_M/a_T}]}$. Its Plummer trial source has mass 6×10^{10} solar masses and scale 3 kpc. Rotation is read at 8.2 kpc. The lensing trial uses impact parameter 10 kpc, finite boundary $R = 200$ kpc and $\Phi = \Psi$. These are reproducible trial-source outputs rather than a new fit to observed galaxies. The stress-source and geometric interpretations read one load and are not added twice.

Table 3 Two explicit physical calibrations on the same calculation rules

Quantity	Inherited energy target	Alternate energy target
Fractions ϕ D R	0.0493 / 0.265 / 0.6857	0.05 / 0.268 / 0.682
u in J m^{-3}	$7.6689477678 \times 10^{-10}$	$7.6689477678 \times 10^{-10}$
P in Pa	$-5.2585974844 \times 10^{-10}$	$-5.2302223777 \times 10^{-10}$
q_0	-0.528550000	-0.523000000
Rotation in km s^{-1}	207.510905127	207.688109769
Conditional deflection in arcsec	0.535586511	0.536727827

The alternate column is consistent with the uniform connection in Physics extension r8 [2]. It does not overwrite the inherited reviewed calibration. Address fractions become equal physical energy fractions only after the declared physical response calibration. The two q_0 values consequently describe different stated inputs. Neither is directly inserted as its own output target.

At fixed volume, the same SI energy operator generates unitary evolution. Noncommuting D and R loads can exchange energy while the total state-change contribution vanishes. During volume change, work and the common pressure enter the homogeneous continuity relation.

$$U(\tau) = e^{-i\hat{E}\tau/\hbar}, \rho(\tau) = U \rho_0 U^\dagger, \quad (14)$$

$$\text{Tr}(\hat{E} \dot{\rho}) = 0, \dot{u} + 3H(u + P) = 0.$$

Address admission also changes system energy. The executed destination-minus-source account has net conversion work $1.3279304601 \times 10^{-12}$ J at completion, with an equal opposite environmental entry. This signed exchange is not a fourth cosmic energy fraction. Recovery S to R has zero

exchange because their per-address response energies agree. The microscopic environmental law producing admission is not derived by this balance.

The Physics extension tests a relative vacuum-reference rule. Adding a common scalar energy term to both normalized states cancels in their relative readout and its volume derivative. The positive R background remains in the gravitational ledger.

$$\begin{aligned} E_g(V) &= \text{Tr}[(\hat{\rho} - \rho_0)H_\mu(V)] + E_R(V), \\ H_\mu &\mapsto H_\mu + C(V)I : \Delta E_g = 0. \end{aligned} \quad (15)$$

The r8 finite tests preserve $E/(u_* V_0) = 1$ and $P/u_* = -0.682$ at the reference volume for common shifts 0, 10^{40} and 10^{120} . A change acting only on the active states remains as real energy and pressure. These archived controls establish the finite relative rule under the given map. Applying the same subtraction to a covariant action, interacting quantum fields and phase transitions remains a connection condition, so full quantum-vacuum closure is not inferred from common-term cancellation.

6 Sequential calibration and residual freedom

The five-anchor calculation permits no future target to affect an earlier stage. G fixes gravitational normalization; adding H_0 fixes the reference density; the phenotype fraction determines hidden load; the clustering fraction separates D from R and enables pressure, q and local response; electron energy supplies the mass unit. Replacing future targets by unavailable values leaves earlier outputs unchanged [5].

The identifiable geometric quantity is $Q_h = \zeta_T \kappa_h^2$. Rescaling stiffness by r and kh by $1/\sqrt{r}$ preserves Qh. The length coefficient $5.746639841098 \times 10^{25}$ m is calculable, but cosmic length L0 requires an additional loop-record magnitude W0. A microscopic cavity length, Hubble length and cosmic size are not interchangeable. A finite periodically glued T^3 is a constructibility choice, not a measured actual topology.

Two response shapes with $\lambda_R = 0$ and 1, separately calibrated to the same reference, both reproduce $q_0 = -0.52855$. Changing K to 4 after that fixation gives q = -0.547819855 and -0.548643149. Thus five-anchor identification and full response-shape identification are different tasks. The calculated local sensitivity rank is five conditional on the fixed structural choices; it does not identify every free parameter. Address controls hold SI coefficients fixed, while changed physical targets require explicit recalibration.

Electron mode 23 and zero intercept are adopted in the MCC branch. Their physical origin is not obtained from the five anchors. With the electron calibration fixed, the energy unit and its time unit are calculated once.

$$\mu_E = E_e / 23, t_\mu = \hbar / \mu_E, \ell_\mu = c t_\mu, \Delta \tau_* = \epsilon_\tau t_\mu. \quad (16)$$

Table 4 Electron units and finite construction choices

Quantity	Value	Role
μ_E	22217.345682174 eV	Calibrated common energy unit
t_μ	$2.962603932832 \times 10^{-20}$ s	Unit reference
ϵ_τ	0.05	Adopted phase interval
$\Delta \tau_*$	$1.481301966416 \times 10^{-21}$ s	Proper-time update spacing

Quantity	Value	Role
Finite mode count	128	Operator support
Cavity length L	$1.776332630208 \times 10^{-10}$ m	Chosen $20 \ell_\mu$ boundary interval

The chosen spacing is not a measured minimum time. Planck time is a comparison scale; $c\Delta\tau_*$ does not prove an independent physical minimum length. Expressing a macroscopic energy operator in electron units changes its units without adding electron energy a second time to cosmic density.

7 Proper time boundaries and allowed spectra

Updates belong to each timelike worldline's accumulated proper time. In the adopted weak static patch $A = 1 + 2\Phi/c^2$ and $B = 1 - 2\Phi/c^2$. Local speed is measured by a static observer, not equated to coordinate speed. The flat limit has $A = B = 1$.

$$\begin{aligned} c^2 d\tau^2 &= A c^2 dt^2 - B dx^2, \\ \frac{d\tau}{dt} &= \sqrt{A(1 - \beta_{local}^2)}, k = \lfloor \tau / \Delta\tau_* \rfloor. \end{aligned} \quad (17)$$

For coordinate duration $T_{coord} = 64 \Delta\tau_*$, flat speeds $\beta = 0, 0.6$ and 0.8 give 64, 51 and 38 complete updates; fractional remainders are retained. The prescribed accelerated path $\beta(s) = 0.2 + 0.5s$ has $\tau / T_{coord} = 0.877980386396$, 57 events including $k = 0$ and 56 completed updates. Chord refinement and a Lorentz-boost control verify the same accumulated proper time. This checks a specified path rather than derives acceleration from coupled gravitational motion. A null path has no assigned rest-clock count; zero null proper time does not freeze photon propagation.

The finite potential is defined by $\Phi(r) = - \int_r^R g(s) ds$ and satisfies $\Phi(R) = 0$ at $R = 200 kpc$. Its

derivative returns the inherited acceleration. At 8.2 kpc the static proper-time ratio is 0.999998751306 and the moving-clock ratio at the reference rotation is 0.999998511748. Uniform FRW clocks use the stationary carrier I/N, retaining the same homogeneous energy and pressure. Enormous clock counts are floating-point estimates, not simulations of every event.

A self-adjoint finite fold operator uses Fourier matrix F , mode numbers -64 through 63 and boundary phase $\eta_B = 0$ or $1/2$. The boundary phase, selection-flow rate η and SI response coefficients η_s are different quantities.

$$H_{fold, \eta_B} = F \text{diag}(\mu_E |n + \eta_B|) F^\dagger, \psi(y+1) = e^{2\pi i \eta_B} \psi(y). \quad (18)$$

A separate spatial positive-energy operator uses the electron rest energy and a disclosed finite interval. Periodic boundaries permit zero momentum; Dirichlet boundaries set both endpoint values to zero. The interval does not represent a physically infinite potential barrier.

$$H_{space} = \sqrt{E_e^2 I + (\hbar c)^2 K}, k_n^{periodic} = 2\pi n / L, k_n^{Dirichlet} = \pi n / L. \quad (19)$$

Table 5 Selected finite spectral outputs without recalibration

Operator and boundary	Mode	Total energy in eV	Kinetic energy in eV
Fold $\eta_B = 0$	23	510998.950690000	Not used
Fold $\eta_B = 1/2$	23	522107.623531087	Not used

Operator and boundary	Mode	Total energy in eV	Kinetic energy in eV
Spatial Dirichlet	1	511010.867747384	11.917057384
Spatial Dirichlet	2	511046.617252180	47.666562180
Spatial periodic	0	510998.950690000	0
Spatial periodic	1	511046.617252180	47.666562180

Boundary changes alter allowed modes under the same calibration. The table is not a fit to atomic lines or an identified observed cavity. Free dispersion $E(p) = \sqrt{E_e^2 + c^2 p^2}$ also executes for noninteger finite real momentum. Discrete update time alone does not discretize every energy or momentum. Equal opposite fold modes do not assign charge signs; charge remains in the independent channel ledger.

The proper-time unitary and its phase-energy branches are disclosed together. Reading an eigenphase alone leaves an integer ambiguity.

$$U_* = e^{-iH\Delta\tau_*/\hbar}, U_* v_j = e^{-i\theta_j} v_j, \\ E_j = \frac{\hbar}{\Delta\tau_*} (\theta_j + 2\pi \ell_j), \ell_j \in \mathbb{Z}. \quad (20)$$

At the default spacing, the maximum energy recovery error from independently computed phases is $6.519258022308 \times 10^{-9}$ eV. At $\epsilon_\tau = 0.25$, 77 fold modes require nonzero branches. Dropping them causes an error of $8.664764816048 \times 10^5$ eV; restoring the declared branches reduces the error to $4.656612873077 \times 10^{-10}$ eV. Branches are supplied from the generator and are not newly inferred from the unitary alone.

8 Connected clocks and the preserved failed identification

The normalized 0.10 collective energy unit $E_* = u_{crit} V_0$ and dimensionless coordinate ξ have an exact fixed-volume conversion to SI proper time. Fold and cavity operators are separate microscopic probes and do not duplicate the collective energy.

$$\exp\left(-i\xi \frac{\hat{E}}{E_*}\right) = \exp\left(-i \frac{\hat{E}\tau}{\hbar}\right), \quad (21) \\ \xi = \frac{E_*\tau}{\hbar}.$$

The executed noncommuting state at $k = 0, 1, 2, 4$ and 8 adopted proper-time intervals preserves total energy $3.200810499959 \times 10^{-10}$ J while the D-load range is 0.038059728227 . State positivity and normalization hold within numerical tolerances. This completes fixed-volume SI phase and internal exchange.

The old nonuniform expansion schedule used $\omega_{info} = 0.7 H_0$. Its frequency is $1.528999668760 \times 10^{-18} \text{ s}^{-1}$, whereas $E_*/\hbar$ is $7.272096256981 \times 10^{24} \text{ s}^{-1}$. Their calculated ratio falsifies identification of those schedules as the same SI phase clock.

$$r_{clock} = \frac{\hbar\omega_{info}}{E_*} = 2.102556972196466 \times 10^{-43} \neq 1. \quad (22)$$

The slow histories retain their status as constitutive-schedule calculations. Uniform I/N remains stationary and permits the homogeneous FRW clock connection. Fully covariant nonuniform carrier,

geometry and expansion dynamics are still open. Substituting a different frequency without deriving its generator would not close that connection.

The local B_C premise adopted in 0.5 places gravity as a geometric response without an independent fundamental graviton Hilbert space. It is a branch assumption, not an experimental proof of the universal impossibility of gravity quantization. Coupling arbitrary matter superpositions to that geometry is part of the unresolved dynamics. The consolidation preserves this premise and the limited executed scope together.

9 Reproducibility and falsification ledger

Clean replay for this consolidation removed captured calculation result directories and regenerated the six 0.10 to 0.12 bilingual sources. It passed 32, 28 and 36 distinct component check groups, respectively. All 37 captured raw result and source files matched the archived release byte for byte. Nine cross-version review contracts also passed. Their document checks refer to the existing reviewed publications; the new consolidated documents have a separate rendering and content audit. Repeated inherited checks are not added again to the distinct count of 96.

The archive is WRRRA_M_0_12_r1_Series_Release.zip at DOI 10.5281/zenodo.23113101. Its current root SHA256 manifest verifies all 312 listed archived files; older version-specific manifests retain historical metadata hashes. The frozen 0.11 input hash is 67de28abf0741c406178a2fe4ce07962e69bce827c3196b78e887da3bd0dab13 and the complete 0.12 input hash is c80742fa4d0e0bdbb94e3390a2cc02da4a535168eaf66fb28f7318ec9b7b8b3d. The endpoint correction commit is b4aa53a481608add2f9b41e432ea516f6bc08fcb.

Physics r8 records additional controls: 33 vacuum-reference checks, 12 uniform-bridge checks and 38 phase-to-load checks, plus 14 inherited phase-frame checks. These are archived supporting results [2]; they are not included in the 96 replay count above. Their common 5% partition connection is preserved without claiming that every microscopic address-to-particle map was executed.

Reproduction commands from the extracted downstream archive are:

```
python calculations/wrra_m_0_10/run_release.py
python calculations/wrra_m_0_12/run_release.py
python review/0_10_to_0_12_r1/review_contracts.py
```

The archive requirements file supplies its numerical dependencies. Word and PDF are publication renderings; generated numerical results and their tests are the computational evidence.

Table 6 Closure ledger and rejection conditions

Connection	Status at downstream closure	Falsification or missing operation
Channel placement and calibrated filter	Executed conditionally	Rank loss charge mismatch nonpositive gap or unsupported winner
Address residue and common partition	Executed	Negative effects repeated admission or inconsistent denominator
SI energy pressure and exchange	Executed	Negative energy derivative mismatch or missing signed conversion work
Uniform gravity and expansion response	Executed conditionally	Same-load disagreement local response failure or duplicate energy

Connection	Status at downstream closure	Falsification or missing operation
Proper-time finite modes and fixed-volume SI phase	Executed	Lorentz clock disagreement spectrum mismatch or lost phase branch
Slow nonuniform schedule as SI phase	Failed identification	Frequency ratio differs from one
Physical outcomes probabilities and post-observation state	Unimplemented	An outcome instrument and apparatus coupling have not been executed
Physical record formation and stability	Unimplemented	No generated record state or record-energy allocation
Repeated-outcome uncertainty	Unimplemented	No repeated measurement ensemble
General covariant dynamics and actual cosmic size	Open	Nonuniform coupled action propagation and geometric inputs required
Actual confinement and microscopic response origin	Open	Boundary selection response shapes and address-channel realization required

Coherent overlap is not an outcome probability or an observation record. The physical record list is empty. A finite spectrum and unitary update therefore do not establish measurement completion. Standard quantum measurement language in a companion paper supplies possible connection conditions, not an executed instrument in this release. The endpoint is the research-series closure with these entries retained.

10 Conclusion

The downstream series now has a single explicit route from channel and address structure to calibrated physical loads, the pressure of the same energy function, conditional geometry and proper-time finite spectra. The route reproduces inherited benchmarks and makes state, boundary and response changes calculable under fixed calibration. Its structural explanation is evaluated through the actual transformations and falsifiers.

The Physics matter and extension manuscripts and the downstream series are consistent when their address measures, energy calibrations, clocks and Actual scopes are kept explicit. Version 1.0 completes the downstream consolidation of the executable 0.1 through 0.12 research scope. It preserves the zero-gap prototype, clock-identification failure and unimplemented measurement and record operations as part of the scientific result.

References

- [1] Choi W, Choi J. Reproduction of the Ordinary Matter 5 Percent Fraction A WRRRA Two Stage Dimensional Filter Model. Korean manuscript r9, 2026. DOI 10.5281/zenodo.23112865.
- [2] Choi W, Choi J. WRRRA Extension Possibilities and Connection Conditions Vacuum Geometry Life and Microscopic Implementation Beyond the Matter Fraction. Korean manuscript r8, 2026. DOI 10.5281/zenodo.23112885.
- [3] Choi W. WRRRA M 0.1 through 0.6 reviewed series. Version 0.6-r2, 2026. DOI 10.5281/zenodo.23076547.
- [4] Choi W. WRRRA M 0.7 through 0.9 reviewed series. Version 0.9-r1, 2026. DOI 10.5281/zenodo.23091892.
- [5] Choi W. WRRRA M 0.10 through 0.12 reviewed series SI Energy and Pressure Sequential Calibration and Proper Time Mode Spectra. Version 0.12-r1, 2026. DOI 10.5281/zenodo.23113101.
- [6] Choi W. Downstream closure at 0.12 and consolidated 1.0 scope. 3 October 2026. https://github.com/Wonsik-Choi-janefather/wrra-m-0.1/blob/b4aa53a481608add2f9b41e432ea516f6bc08fcb/DOWNSTREAM_CLOSURE_0_12.md.
- [7] Choi W. WRRRA Core 1.0 A Domain Agnostic Execution Architecture. Frozen edition, 2026. Source lineage preserved in the downstream series [3-5].
- [8] Choi W. Minimal Computation Cosmology 2.3.2. Frozen commit 21daec110c0cbeeb228c445d587eac8302e7f767, 2026. <https://github.com/Wonsik-Choi-janefather/minimal-computing-cosmology-2.3.2>.

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